

Local Outlier Factor as a Third Detector

Round 5 experiment report — multivariate time-series anomaly detection (batch-distillation sensor data)

1. Executive summary

We integrated a teammate's Local Outlier Factor (LOF) detector as a third model and tested whether fusing it with our Round-4 CNN autoencoder could beat the CNN alone. Using leakage-free, run-grouped cross-validation to choose the fusion weight, and testing LOF both on raw and on standardized features, the answer was a consistent **no**: the cross-validation assigned LOF essentially zero weight, and the honest hybrid never exceeded the CNN alone. **CNN-alone remains the model of record.**

This is a genuine, well-supported negative result. LOF is a sound detector and the integration is correct; it simply cannot out-argue the Round-4 CNN under our precision-weighted score.

2. What LOF is, and why it was worth trying

Local Outlier Factor is a **density-based** anomaly detector. For each point it finds its k nearest neighbours and measures how tightly packed that local neighbourhood is, then compares the point's local density to its neighbours' densities. A point sitting in a much sparser pocket than the points around it receives a high LOF score. Crucially, the judgement is *local*: a point is flagged for being lonely relative to its own neighbourhood, not relative to the whole dataset.

Why a third detector at all? Diversity.

An ensemble only gains when its members make *different* kinds of mistakes. Our three detectors have genuinely different inductive biases, so in principle they can cover each other's blind spots:

- **CNN autoencoder** — reconstructs normal temporal windows; flags what does not reconstruct like normal data (temporal/shape view).
- **Isolation Forest** — global, tree-based; flags points that are easy to isolate with random splits (global-rarity view).
- **LOF** — local, density-based; flags points in locally sparse regions, even if they look unremarkable globally (local-density view).

Standalone, LOF (val F1 $\approx$ 0.46–0.48) also outscored our Isolation Forest baseline (F1 $\approx$ 0.49 at high recall / low precision), so it was the more promising tabular partner to try. The hypothesis was reasonable: a density-based detector might catch anomalies the reconstruction-based CNN misses.

3. What we did

The teammate's fusion had been built against the pre-Round-4 CNN (flat MSE scoring, plain F1). We did not merge it blindly; we cherry-picked the ideas onto the current codebase and rebuilt the pipeline (`src/pipelines/run_final_hybrid.py`):

- **Kept the Round-4 CNN unchanged** — Mahalanobis distance over per-sensor reconstruction error, F-beta = 0.5 (precision-weighted) threshold, best-checkpoint selection.
- **Adopted the cross-sensor feature set** — per-sensor lag / delta / rolling / EWMA plus pairwise products, differences, and rolling correlations (~456 features), which capture sensors decoupling from one another.
- **Dropped Isolation Forest from the fused model** — it had taken ~0 weight in every prior sweep.

- **Rigorous weight selection** — one free parameter (`cnn_weight`); chosen by run-grouped 5-fold cross-validation so no run leaks between folds; the deployed weight is the *mean* of the per-fold weights, and we report the honest out-of-fold (OOF) hybrid score.
- **Two experiments** — (A) LOF on the raw features, then (B) LOF on features standardized on training statistics only, to stop the large-magnitude product columns from dominating LOF's distance metric.

Both experiments were run on the university V100 GPU cluster (CNN on GPU; LOF is CPU-bound). All numbers below are from those authoritative cluster runs.

4. What happened (results)

Experiment A — LOF on raw features

Model (validation)	F1	ROC-AUC	LOF weight
CNN only	0.592	0.783	&mdash;
LOF only	0.461	0.687	—
Hybrid (honest OOF)	0.549	0.716	0.03
Hybrid (avg weight, full val)	0.564	0.761	0.03

Per-fold chosen CNN weight: [1.00, 1.00, 1.00, 0.85, 1.00] → deployed LOF weight 0.03.

Experiment B — LOF on standardized features

Model (validation)	F1	ROC-AUC	LOF weight
CNN only	0.568	0.765	&mdash;
LOF only	0.481	0.667	—
Hybrid (honest OOF)	0.568	0.765	0.00

Per-fold chosen CNN weight: [1.00, 1.00, 1.00, 1.00, 1.00] → deployed LOF weight 0.00. With zero weight the hybrid is identical to the CNN. (CNN-only differs from Experiment A only by ordinary run-to-run checkpoint variance.)

5. Why fusion did not help

- **The cross-validation rejected LOF.** In 9 of 10 folds across both experiments, the weight search independently chose pure CNN. This is the most trustworthy signal — it is leakage-free.
- **Even a small LOF blend hurt.** In Experiment A a 3% LOF mix lowered the honest OOF F1 from 0.592 to 0.549: recall rose but precision fell sharply, and under precision-weighted F-beta = 0.5 that trade is a net loss.
- **Standardizing did not rescue it.** LOF's F1 nudged up (0.46 → 0.48), but its ROC-AUC actually fell (0.687 → 0.667) — its *ranking quality*, which is what fusion needs, did not improve; the F1 change was only a shift to a higher-recall operating point.
- **The CNN is simply stronger.** CNN ROC-AUC (~0.78) clearly exceeds LOF's (~0.67). Mixing a weaker, differently-distributed score into a cleaner ranking corrupts it rather than complementing it.

6. Conclusion and recommendation

Across three fusion attempts — Isolation Forest, raw-feature LOF, and standardized-feature LOF — no tabular detector has been able to lift the Round-4 Mahalanobis CNN under precision-weighted scoring. We therefore adopt **CNN-alone** as the final model, and treat the fusion investigation as a rigorously established negative result rather than a gap.

The LOF work is not wasted: it provides the evidence that fusion was tried carefully and correctly rejected, and the run-grouped CV methodology is a reusable, honest tool. The most promising remaining lever targets the CNN itself (e.g. weight decay, early stopping, and robust down-weighting of likely-contaminating training windows), not further fusion.

Recommendation: submit the standalone CNN (outputs/cnn_predictions.csv, val F1 $\approx$ 0.61) to the portal; do not submit any hybrid file.